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Ms Claudia Schlüter
Progroup AG
Sandersdorf-Brehna, Germany

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Application for Betriebsingenieur KI-Anwendungen & Prozessmodellierung

Dear Ms Schlüter,

I am writing to apply for the Betriebsingenieur KI-Anwendungen & Prozessmodellierung position at Progroup. I have a Master's in Artificial Intelligence and three years of experience building and deploying AI applications, and the part of your posting that fits me best is turning process data into models and assistants that people on the shop floor can use.

At Covestro I worked as a data scientist on a chemical production process. I built Random Forest, XGBoost and LSTM models that predict viscosity and reaction progression from sensor time series, engineered lag and rolling-window features, and compared them with empirical curve fitting. The result helped to spot batches that would drift from specification earlier. I also built a Streamlit dashboard so chemists could change parameters and see the predicted outcome before running a physical batch. I would like to do that kind of work for paper machines, chemicals and water circuits.

Your "Fragen statt Suchen" assistants are close to what I do now. At Kniit I am the only developer of a knowledge-extraction system for the German energy sector. Multi-agent extraction with LangGraph raised precision and recall from 33% and 45% to 93% and 90%, and a query service lets experts retrieve answers in milliseconds. In my Covestro thesis I evaluated RAG systems against ground truth from chemists, so I am used to checking an assistant against the people who know the plant. I work in Python and with databases such as Postgres, MongoDB and Neo4j, and I deploy with Docker on AWS and Azure.

There are gaps I should name. I have not worked in the paper or pulp industry, I do not have several years in a process-industry plant function, and I have not written control concepts with automation teams. I have not worked with design of experiments in production either. I would need to learn those from your colleagues in production and automation, and I am happy to do that on site.

Progroup runs three paper mills and fourteen corrugated plants, and a model that works at one site can pay off at the others. Rolling out solutions across locations together with production, maintenance and IT is the part of the role I find most interesting.

I am based in Halle (Saale), hold an EU Blue Card and can start on 15 November 2026. I am fully willing to travel between sites. My German is at B1 level, so I would write documents in English at first. Thank you for considering my application.

Kind regards,

Chandu Prasad Bhairapu